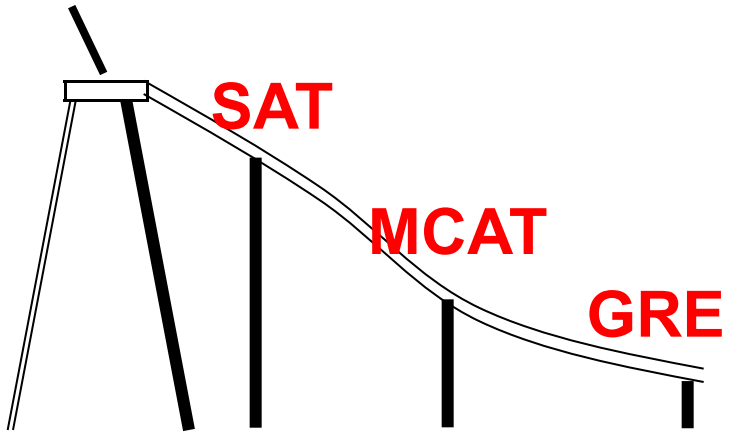
Test Slide



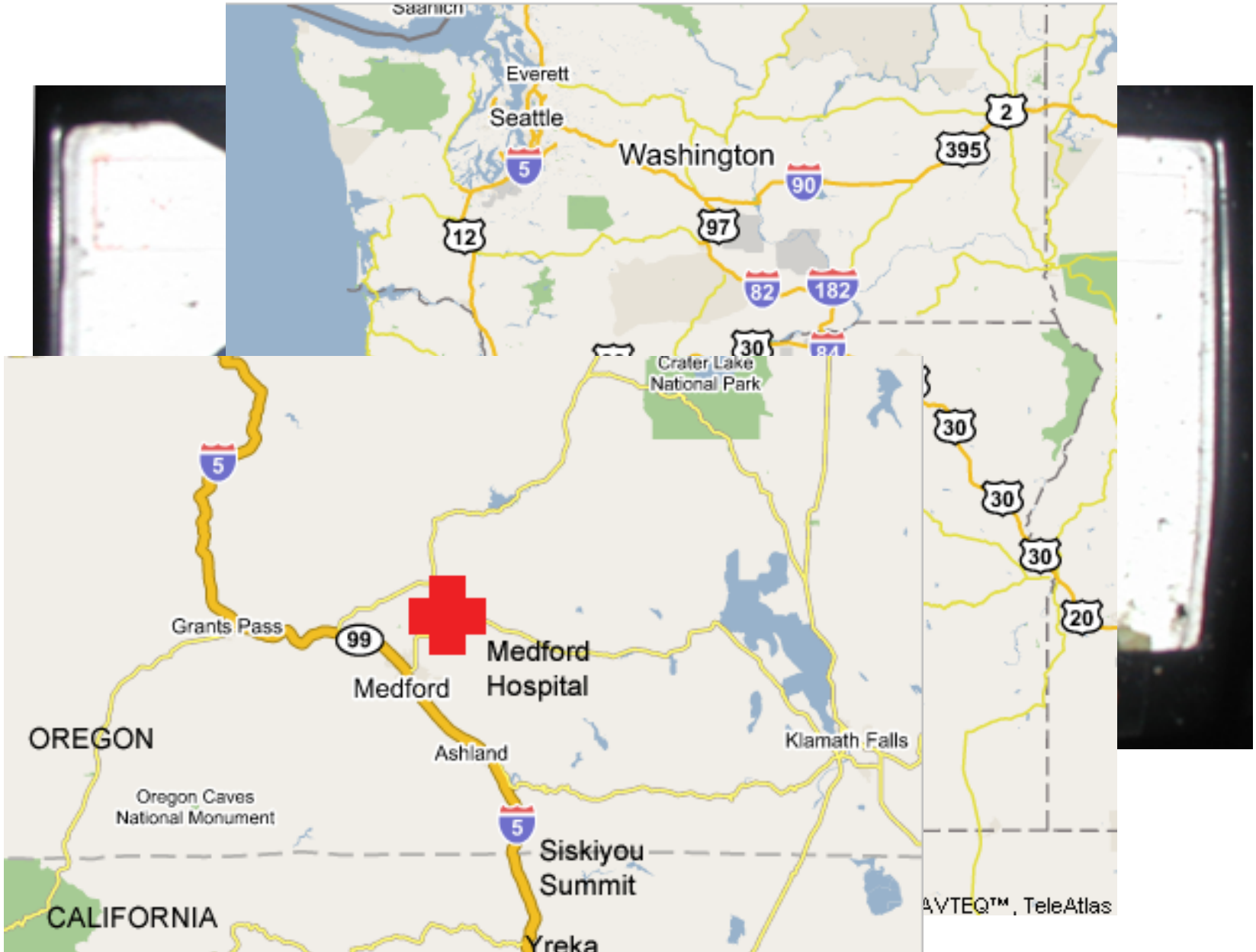
How and What to Quantum Compute



Dave Bacon
Santa Fe Institute

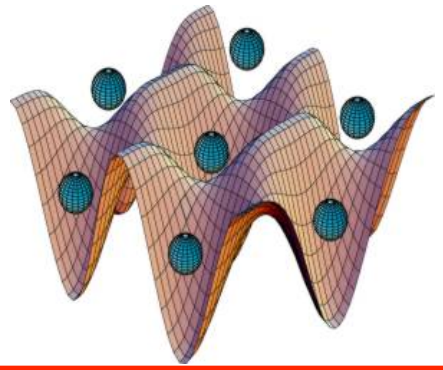
[University of Washington CSE Colloquium, 2/24/05]

All Apologies



20th Century Reduced Rec





Quantum Computation

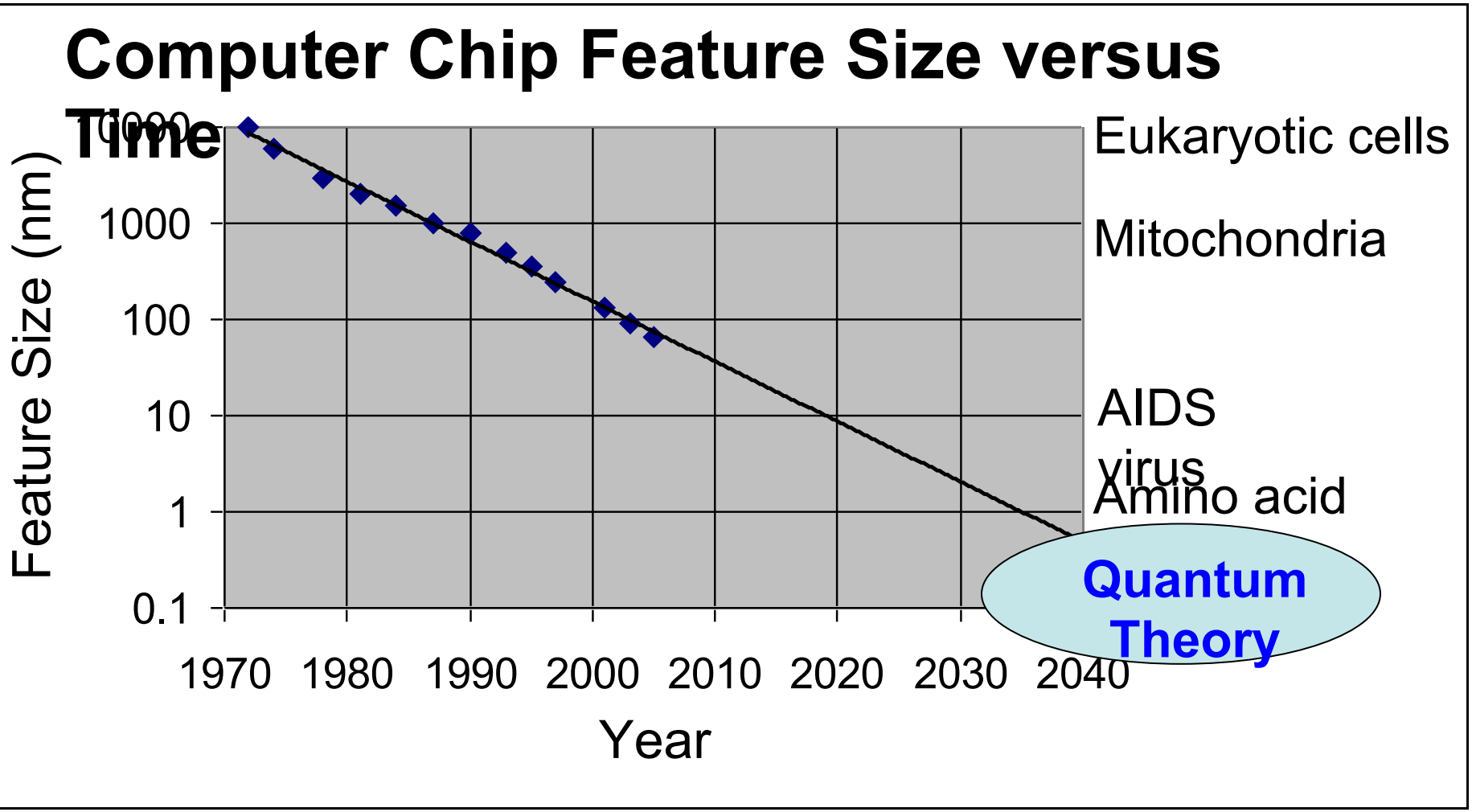


Computation **Physics**

exponential growth of digital computing power combined with the rise of software

elucidation and verification of laws describing how nature operates

Into the Future



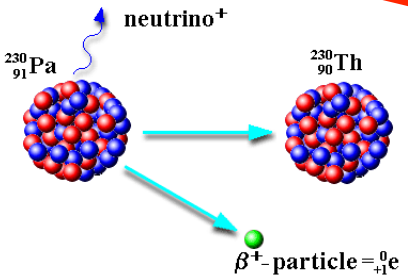
"Prediction is very difficult, especially if it's about the future."



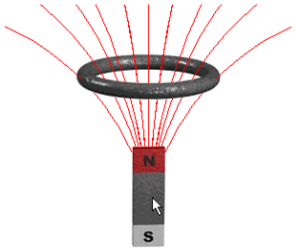
Niels Bohr

What is Quantum Theory?

Weak Force

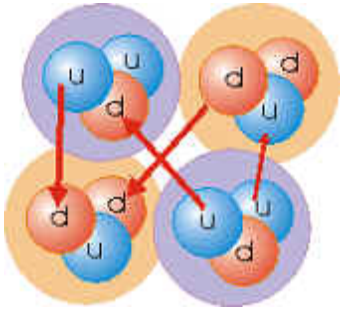


Electromagnetic Force

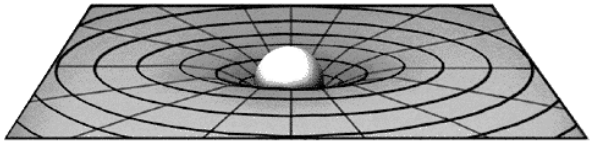




Strong Force



Gravitational Force (?!?!)



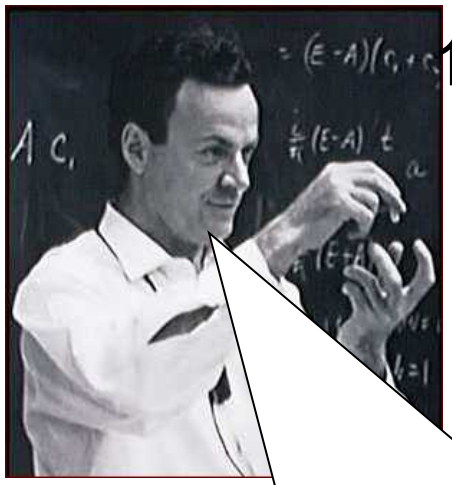
Quantum Theory

Quantum theory is the **machine language** of the universe

A Worrying Physicist

2040 is a long time in the future….so why worry?

Richard Feynman



Observations

1. No efficient computer simulations of quantum theory
2. Systems evolving according to quantum theory efficiently compute properties of the quantum system (that's what experiments are!)

"How can we simulate quantum mechanics?... Can we do it with a new kind of computer…a **quantum computer**? It's not a Turing machine but a machine of a different kind" (1981)

Shocking scoveries



Umesh Vazirani (U.C. Berkeley)

Quantum computers offer superpolynomial speedups (for certain oracle promise problems) over classical computers
[Bernstein, Vazirani STOC 93]

Quantum computers offer exponential speedups (for certain oracle promise problems) over classical computers
[Simon FOCS 94]



Dan Simon (Microsoft)



Peter Shor (AT&T→MIT)

Quantum computers can efficiently factor integers and can efficiently compute the discrete logarithm
[Shor FOCS 94]

The Factoring Firestorm

1881988129206079638386972394616 5043
9807163563379413827007633564229888
5971523466548531906060650474304 5317
3880113033967161996923212057340 3187
9550656996221305168759307650257059

=

39807508642406493739712550055038649119906436234252670840638518957594638895726176858331 7

×

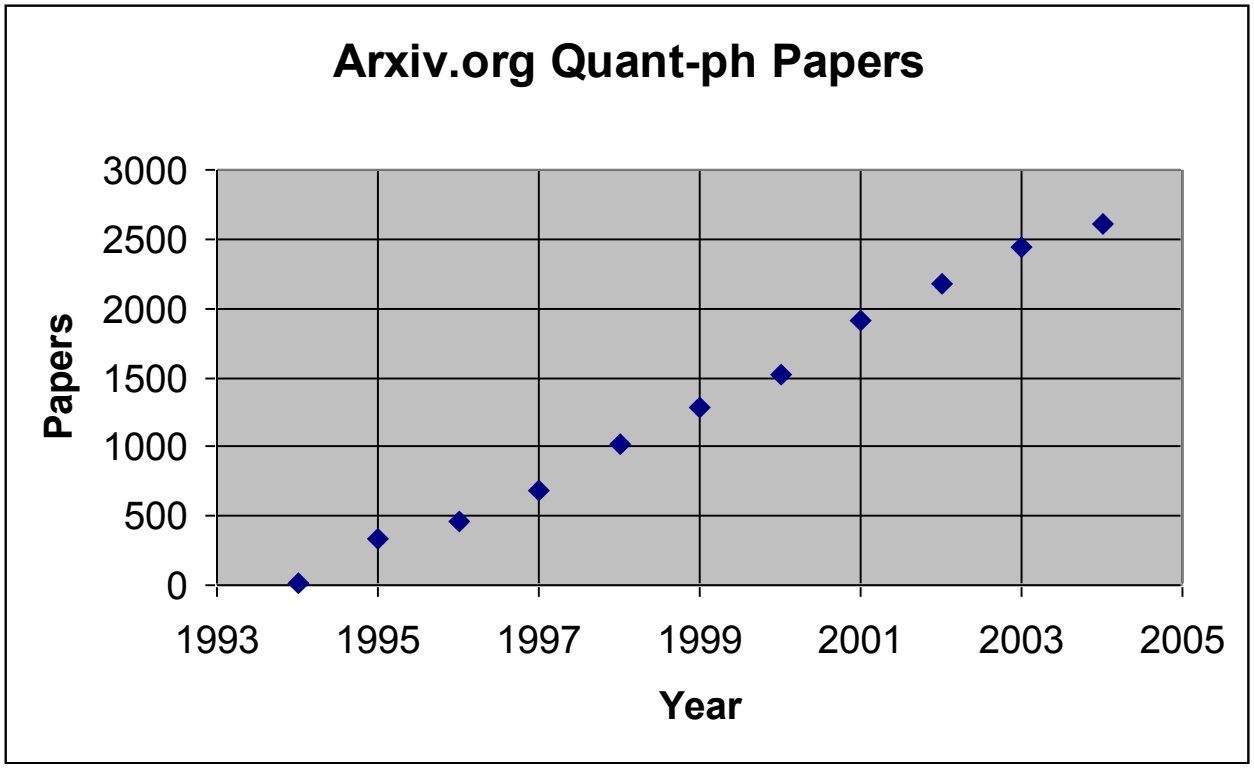
47277214610743530253622307197304822463291469530209711645985217113052071125636359039752 7

Best classical algorithm takes time $O(\exp(n^{1/3}))$

Shor's quantum algorithm takes time $O(n^2 \log n)$

An efficient algorithm for factoring **breaks** the RSA public key cryptosystem

Boom!



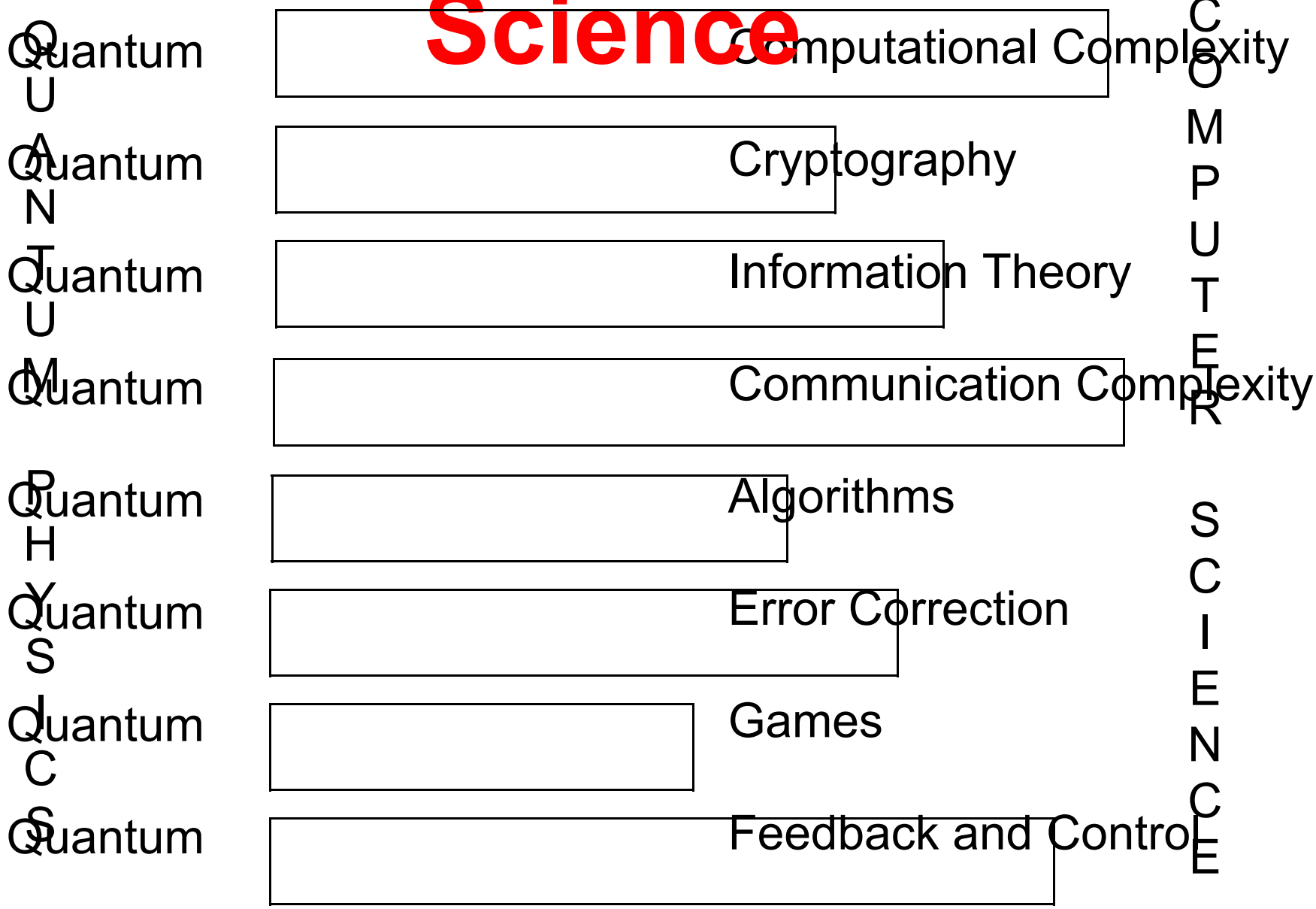
Comparison

2004 quant-ph: 2606
2004 hep-th: 3357
(string theory)
2004 cond-mat: 8961

Quantum Information Science

Every single information processing task should be reexamined in light of quantum theory

Quantum Information Science



Quantum Information Science

INFORMATION PROCESSING

PHYSICS

Quantum Computation in Ion Traps

Quantum Computation in Optical Lattices

Quantum Computation in Quantum Dots

Quantum Computation in Nuclear Magnetic Resonance

Quantum Computation in Superconductivity

Quantum Computation in Spins in Semiconductors

Quantum Computation in Electrons on Liquid Helium

Quantum Computation in Optical Systems

Quantum Computation in Cavity QED

Crossbreeding Results!



Holy Grails of Quantum Information Science

- **Exponential** reduction in quantum resources over classical (time, space, communication cost, etc.)
- **Impossibility** of classical task possible with quantum theory (cryptographic protocols, games, etc.)
- **Methods** of building quantum computers which are robust to noise and component faults (error correction, fault-tolerance, decoherence-reduction)
- **Experimental** realizations of quantum evolution (physical manipulation of quantum information)

Yesterday, Today, Tomorrow

What gives quantum computers their **power** over classical computers

How to jumpstart the **building** of a quantum computer?

Quantum Theory Made Easy

Classical				Quantum			
0	0	0	p0	a0			
0	0	1	p1	a1			
0	1	0	p2	a2			
0	1	1	p3	a3			
1	0	0	p4	a4			
1	0	1	p5	a5			
1	1	0	p6	a6			
1	1	1	p7	a7			

<u>probabilitie</u>

$p_i \in \mathbb{R}$

$0 \leq p_i \leq 1$

$\sum_i p_i = 1$

probability of observing state *s* is *ps*

<u>amplitude</u>

$a_i \in \mathbb{C}$

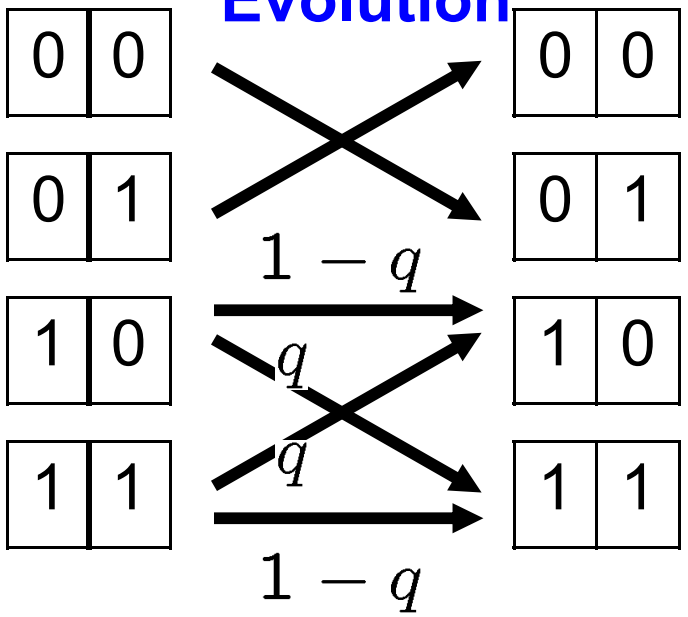
$0 \leq |a_i|^2 \leq 1$

$\sum_i |a_i|^2 = 1$

probability of observing state *s* is |*as*|2

Quantum Theory Made Easy

Classical Evolution

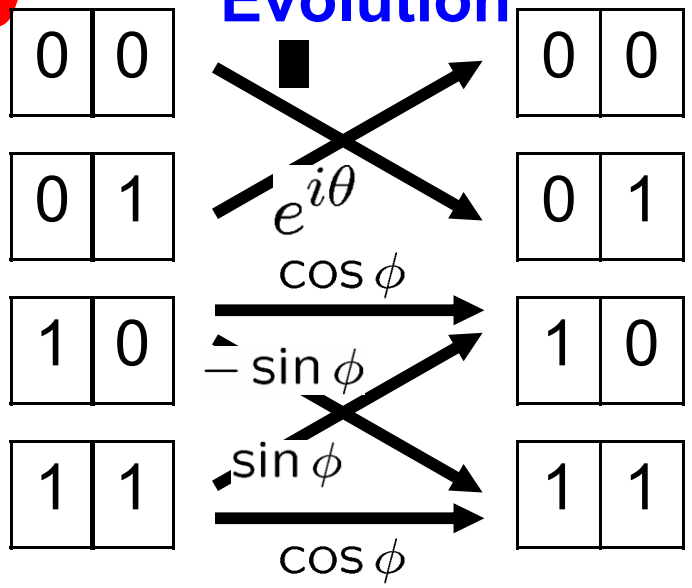


transition probabilities

$$\begin{bmatrix} p'_0 \\ p'_1 \\ p'_2 \\ p'_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1-p & q \\ 0 & 0 & p & 1-q \end{bmatrix} \begin{bmatrix} p_0 \\ p_1 \\ p_2 \\ p_3 \end{bmatrix}$$

stochastic matrix

Quantum Evolution



transition amplitudes

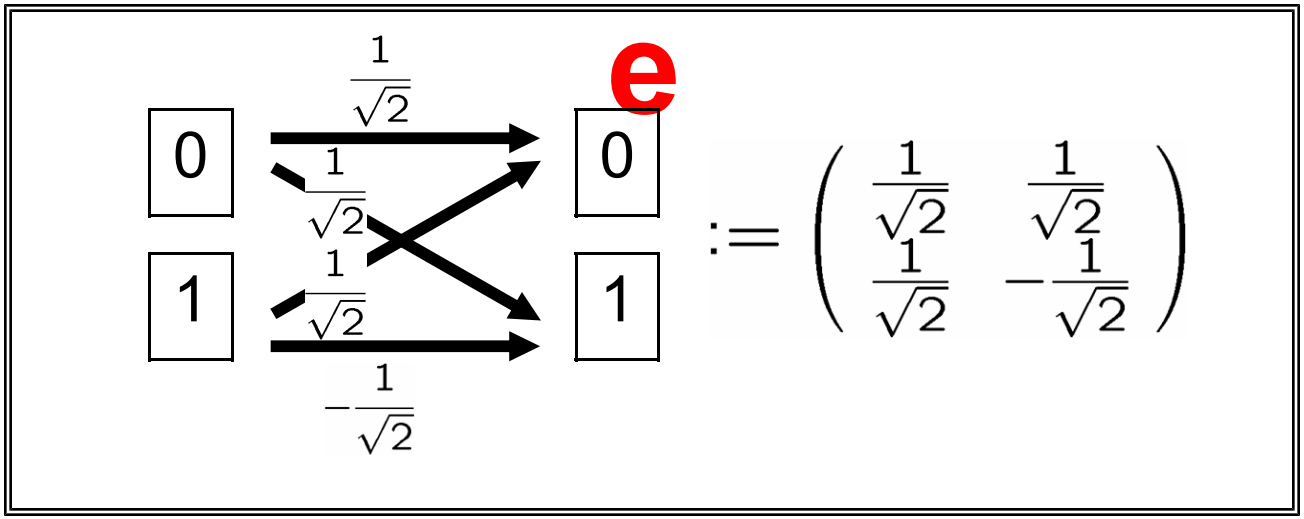
$$\begin{bmatrix} a'_0 \\ a'_1 \\ a'_2 \\ a'_3 \end{bmatrix} = \begin{bmatrix} 0 & e^{i\theta} & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & \cos\phi & \sin\phi \\ 0 & 0 & -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

unitary matrix



Interferenc e





qubit input

$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

0	50 %	100%
1	50 %	0%

$\begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$ — $\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$ — measure

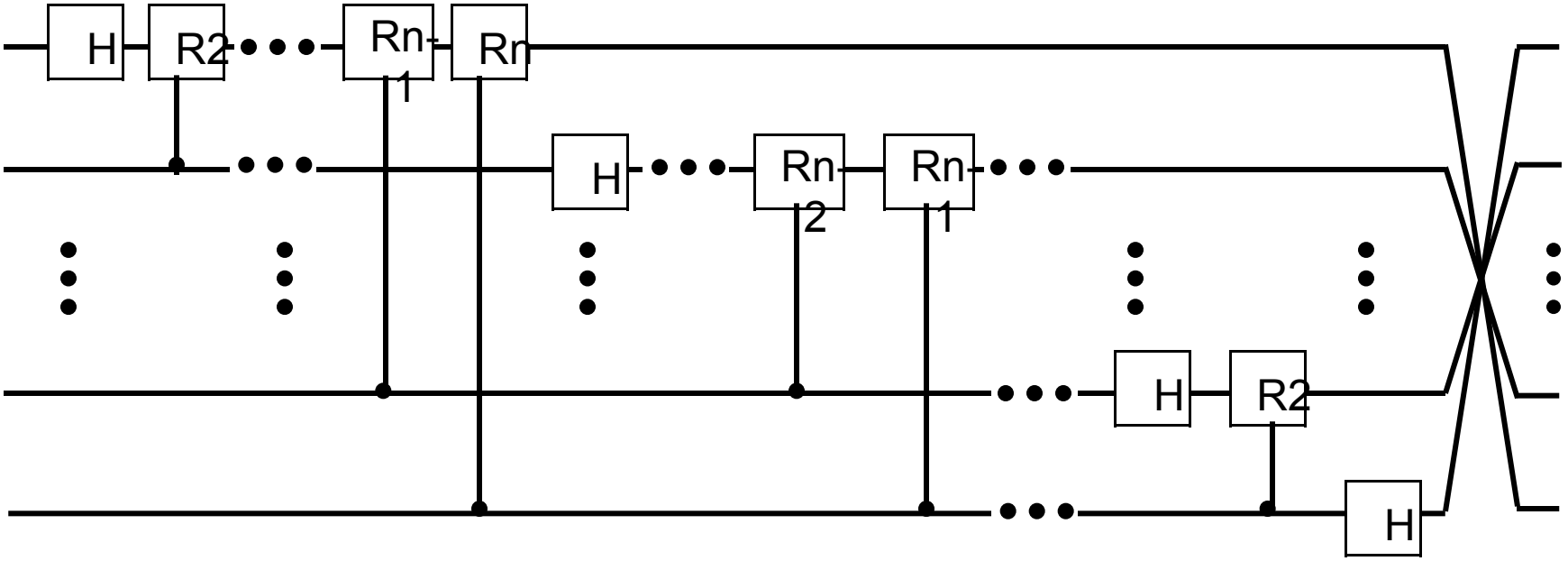
0	50 %	0%
1	50 %	100%

Quantum Algorithms

uniform quantum circuit families

sequence of

- one and two qubit unitary evolutions
- measurements



Quantum circuit for the Fourier Transform over $\mathbb{Z}_{2^n}$

Quantum Algorithms

- *Factoring, Discrete log [Shor 94]
- Unstructured search [Grover 96]
- *Abelian hidden subgroup problems [Shor 94, Kitaev 95]
- *Pell's equation [Hallgren 02]
- *Hidden shift problems [van Dam, Hallgren, Ip 03]
- *Graph traversal [CCDFGS 03]
- Spatial search [AA 03, CG 03/04, AKR 04]
- Element distinctness [Ambainis 03]
- Various graph problems [DHHM 04, MSS 03,…]
- Testing matrix multiplication [Buhrman,Špalek 04]
- …[Your Algorithm Here]

Where is the power of quantum algorithms coming from?

* = suspected exponential gain over classical

Minimizing the Yakking



ALICE

a

Feq(a,b) =

1 if a=b

0 if a≠b



BOB

b

0	1	0	1	•••	0

n bits

0	1	1	1	•••	0

n bits

How many bits do Alice and Bob need to exchange (in the worst case) to compute F(a,b)?

With no error: n+1 bits

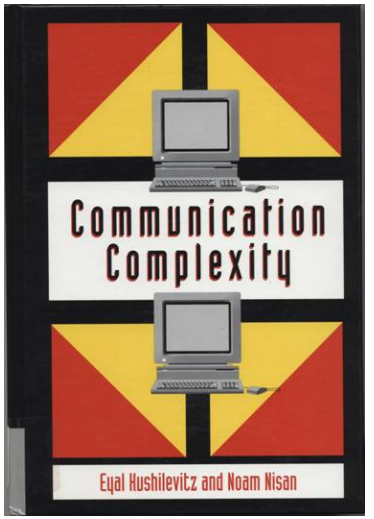
With one-sided error ε: O(log(n/ε)) bits

Communication Complexity



Classical communication costs of distributed classical computations

[Yao, STOC 79]

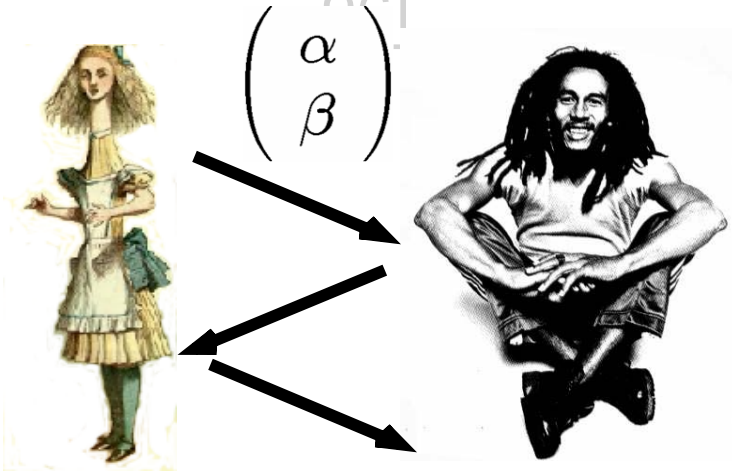


[Kushilevitz and Nisan



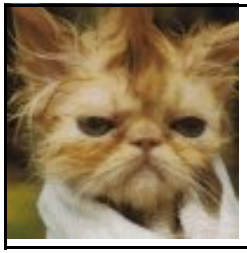
Quantum communication costs of distributed classical computations

[Yao, FOCS 93]



exchange qubits

Quantum Communication Complexity



Holevo's Theoerem: sending n classical bits over a quantum channel requires n qubits



[Raz, STOC 99]

There exist functions *F(a,b)* which require $\Omega(m1/4/\log(m))$ bits of communication to solve
(with error) but can be solved using only $O(\log(m))$ qubits of communication.
(*m* is the length of *a* and *b*)



Exponential benefit of quantum over classical.

Teleportation (Demystified)



Alice wants to send $\begin{bmatrix} p \\ 1-p \end{bmatrix}$ to Bob

$\begin{bmatrix} p \\ 1-p \end{bmatrix}$



1
bit



A

1
bit

B



$\begin{bmatrix} p \\ 1-p \end{bmatrix}$

AB

$\begin{bmatrix} \frac{1}{2} \\ 0 \\ 0 \\ \frac{1}{2} \end{bmatrix}$ 00 01 10 11

$\begin{bmatrix} p \\ 1-p \end{bmatrix}$

If Alice's bit is 1, then flip A

If received bit is 1, then flip B

Quantum Teleportation

Alice wants to send $|\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ to Bob





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

1 qubit

X

Y

2 bits





$\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$ ○ $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}$ ○ $\begin{bmatrix} \alpha \\ \beta \end{bmatrix}$

Alice performs operation on her state and half of entangled system and sends outcome

Conditional on received bits, Bob performs manipulation of his half of entangled system

Rehas



[Raz, STOC 99]

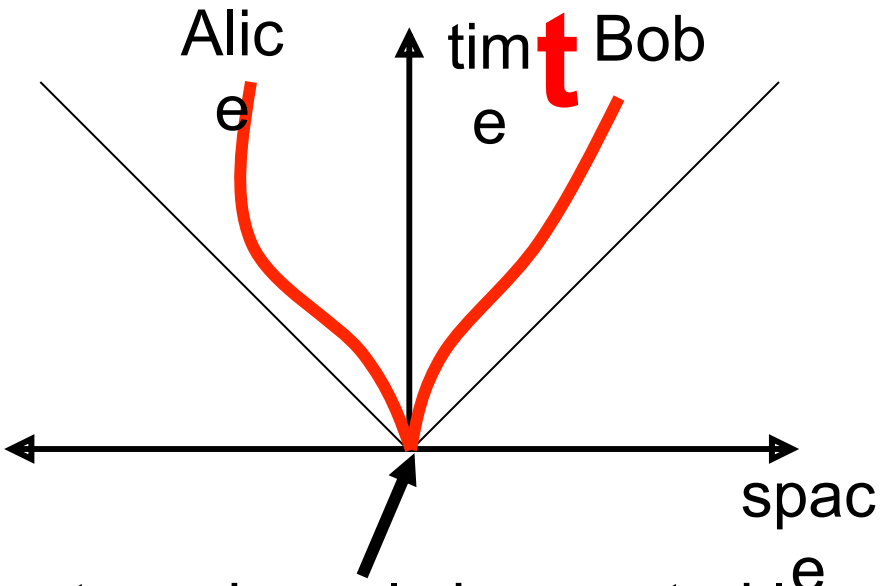
There exist functions *F(a,b)* which require *Ω*(*m*1/4/log(*m*)) bits of communication to solve
(with error) but can be solved using only *O*(log(*m*)) bits and *O*(log(*m*)) ebits of communication. (*m* is the length of *a* and *b*)

1 ebit := $|\phi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

1. Entanglement can be used to gain exponential reduction in communication resources
2. Entanglement is “like” classically correlated bits…but it’s different…

Preliminary thesis: entanglement is involved in quantum benefits

Entanglemen



Entanglement can be only be created locally

…but classical correlations can only be created locally as well

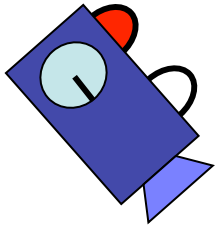
What is the difference between classical correlations and quantum correlations?

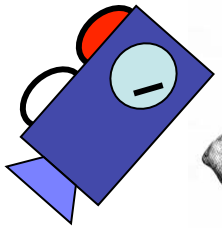
Compar

e

two quantum subsystems









ALICE

3. Choose measurement
4. Measure
5. Get outcome

2. Separate
1. Interact and entangle

3. Choose measurement
4. Measure
5. Get outcome

BOB



ALICE

2. Choose measurement

1. *Share classical correlations*

2. Choose measurement



BOB

3. Produce measurement outcome

data

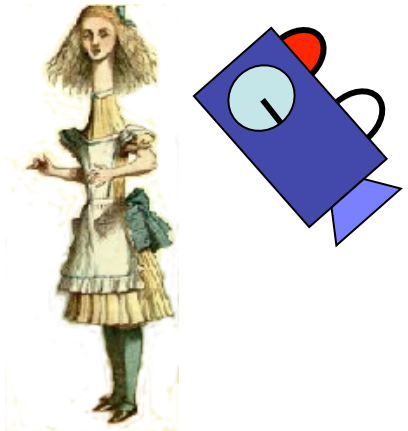
03XSF
W3331

data

03XSF
GF111

3. Produce measurement outcome

An Experiment

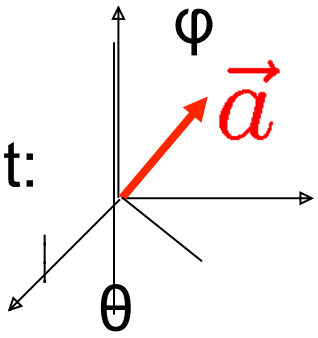




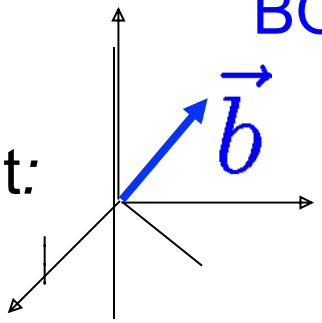
ALICE

BOB

Measurement:



Measurement:



Outcome : $A \in \{+1,-1\}$

Outcome : $B \in \{+1,-1\}$

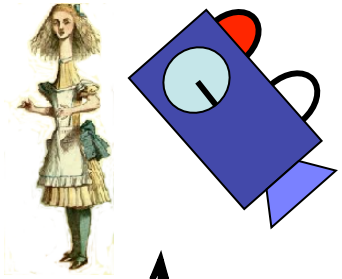
Quantum Experiment Results:

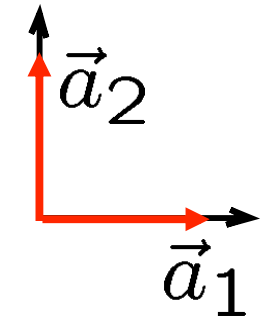
$$\Pr\left[A, B|\vec{a}, \vec{b}\right] = \frac{1}{4}\left[1 - AB\vec{a}\cdot\vec{b}\right]$$

$$\langle A\rangle = \langle B\rangle = 0$$

$$\langle AB\rangle = -\vec{a}\cdot\vec{b}$$

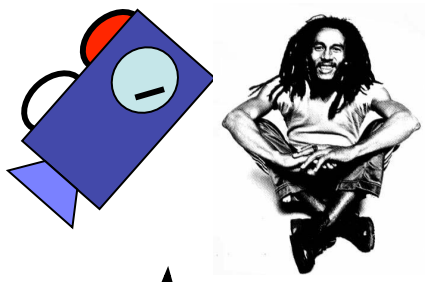
Bell's Theorem

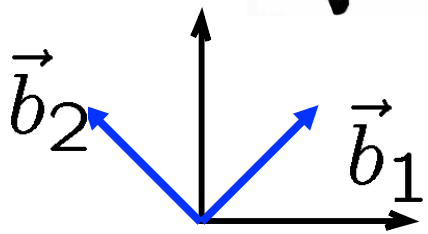




Four Quantum

$$\langle A_1 B_1 \rangle_{QM} = -\frac{1}{\sqrt{2}}$$
$$\langle A_1 B_2 \rangle_{QM} = \frac{1}{\sqrt{2}}$$
$$\langle A_2 B_1 \rangle_{QM} = -\frac{1}{\sqrt{2}}$$
$$\langle A_2 B_2 \rangle_{QM} = -\frac{1}{\sqrt{2}}$$





Four Classical Experiments

A_i cannot depend on B_j (either outcome or which measurement)

$$|A_1 B_1 - A_1 B_2 + A_2 B_1 + A_2 B_2|$$
$$|A_1 (B_1 - B_2) + A_2 (B_1 + B_2)| \leq 2$$
$$|\langle A_1 B_1 \rangle_C - \langle A_1 B_2 \rangle_C + \langle A_2 B_1 \rangle_C + \langle A_2 B_2 \rangle_C| \leq 2$$

Bell's Theorem: quantum theory produces correlations which cannot be explained as locally created classical correlations:

$$|\langle A_1 B_1 \rangle_{QM} - \langle A_1 B_2 \rangle_{QM} + \langle A_2 B_1 \rangle_{QM} + \langle A_2 B_2 \rangle_{QM}| = 2\sqrt{2}$$

A History of Worry



Albert Einstein



Niels Bohr



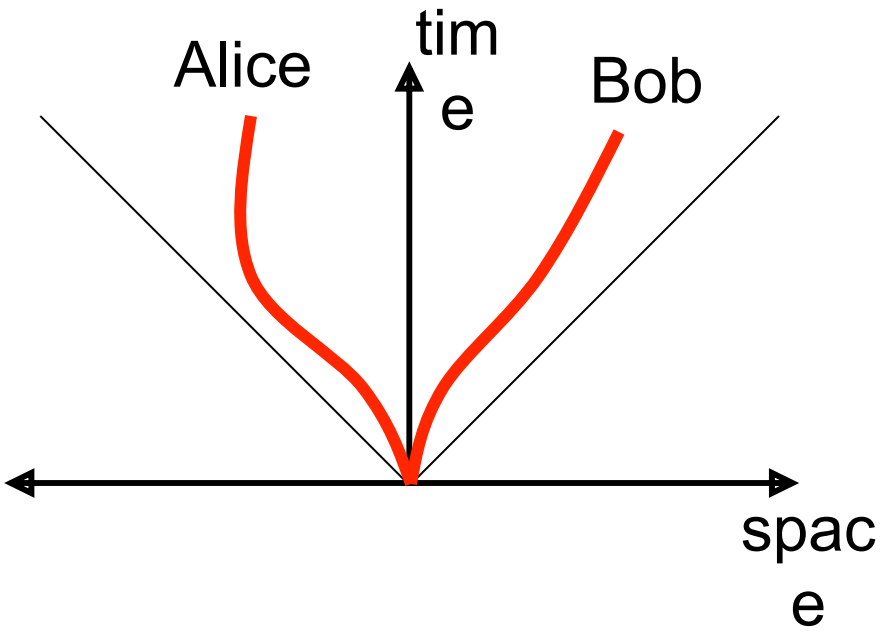
Erwin Schrödinger



Alan Turing



John von Neumann



If Alice and Bob cannot communicate, how can quantum correlations be explained?

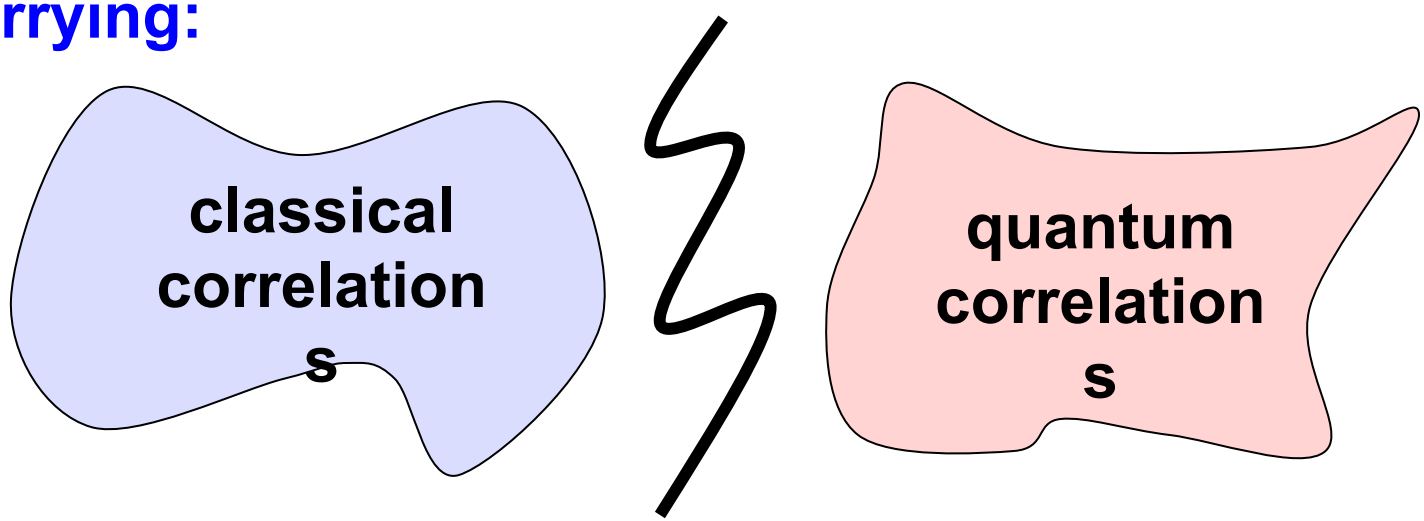
Quantum Information Science



…is what happens when we learn to stop worrying about quantum theory and love the strangeness



Worrying:



Bell's Theorem

Accepting: quantum correlations are a resource

How do we **quantify** this resource?

A New Question

What is the communication cost of simulating the correlations produced by measuring entangled quantum states?

[Bacon and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
[Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]



ALICE

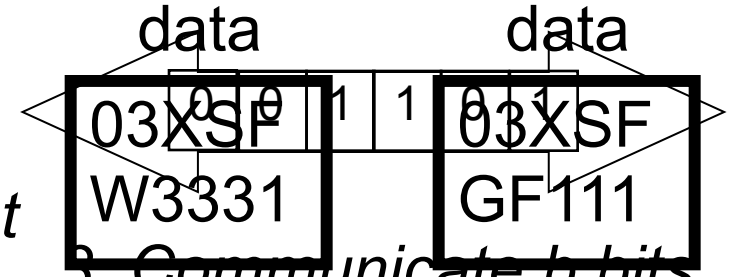
1. *Share classical correlations*

2. Choose measurement

2. Choose measurement



BOB



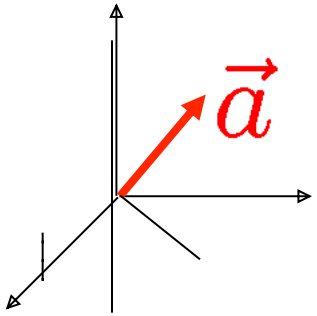
3. Communicate b bits

4. Produce measurement outcome

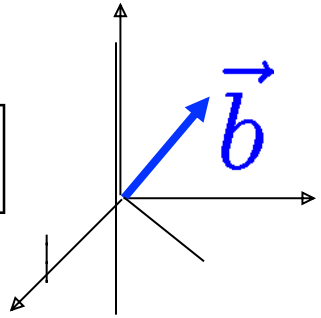
4. Produce measurement outcome

Communication complexity of Nature!

A Protocol



$$\Pr\left[A, B|\vec{a}, \vec{b}\right] = \frac{1}{4}\left[1 - AB\vec{a}\cdot\vec{b}\right]$$



How to simulate without sending full description of vectors between Alice and Bob?

Alice and Bob share two perfectly correlated uniformly distributed unit three dimensional vectors $\vec{\lambda}_1,\ \vec{\lambda}_2$

1. Alice outpu$-\text{sgn}(\vec{\lambda}_1 \cdot \vec{a})$
2. Alice sends$c = \text{sgn}(\vec{\lambda}_1 \cdot \vec{a})\text{sgn}(\vec{\lambda}_2 \cdot \vec{a})$ to Bob
3. Bob output$\text{sgn}((\vec{\lambda}_1 + c\vec{\lambda}_2) \cdot \vec{b})$

Correlations on a single ebit
=
shared randomness + 1 bit of communication

[Bacon and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
[Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]

Finiteness Conjecture

Conjecture: all multipartite entangled systems can be exactly simulated by augmenting a classical correlations with an amount of communication polynomial in the product of the dimensions of the entangled subsystems.

- projective measurements on two entangled qubits
  [Toner and Bacon, *Phys. Rev. Lett.* 91, 187904 (2003)]
  [Bacon and Toner, *Phys. Rev. Lett.* 90, 157904 (2003)]
- all bipartite two outcome correlation observables
  [Bacon and Toner, in preparation (2005)]
- Pauli measurements on a n-party GHZ states
  [Tessier, Caves, Deutsch, Eastin, Bacon, quant-ph/ 0407133 (2005)]
- average case with projective measurements on n ebits
  [Massar, Bacon, Cerf, Cleve, *Phys. Rev. A* 63, 052305 (2001)]
- ....

Back To Algorithms

Entanglement and **Quantum Speedups**:

small quantity of entanglement → efficient classical algorithm for simulating the system

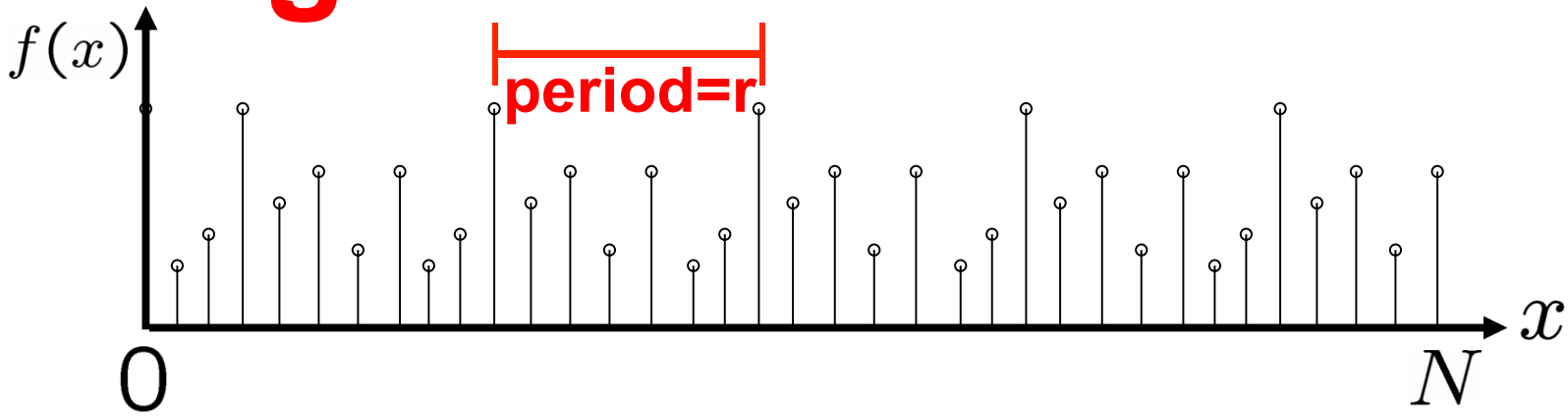
[Jozsa and Linden, Proc. Math. Phys. & Eng. Sci. 459 2011 (2003)]
[Vidal, Phys. Rev. Lett. 93, 040502 (2004)]

How can this help us guide finding new quantum algorithms?

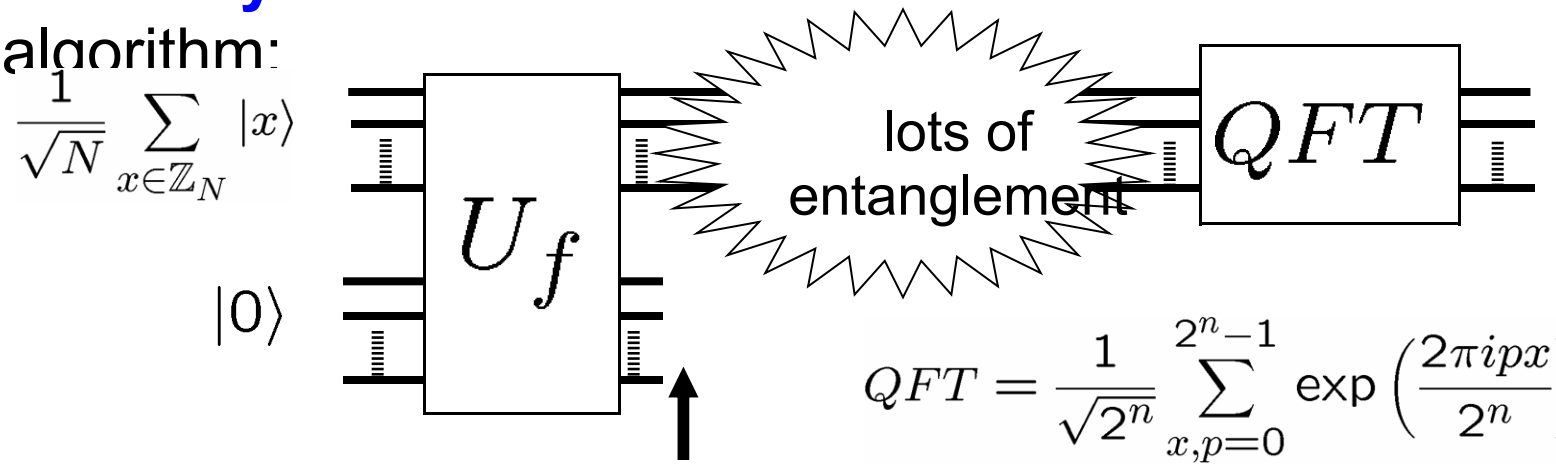
Next: a simple example

Factoring and Entanglement

Factoring can be reduced to period finding.



Anatomy of Shor's algorithm:

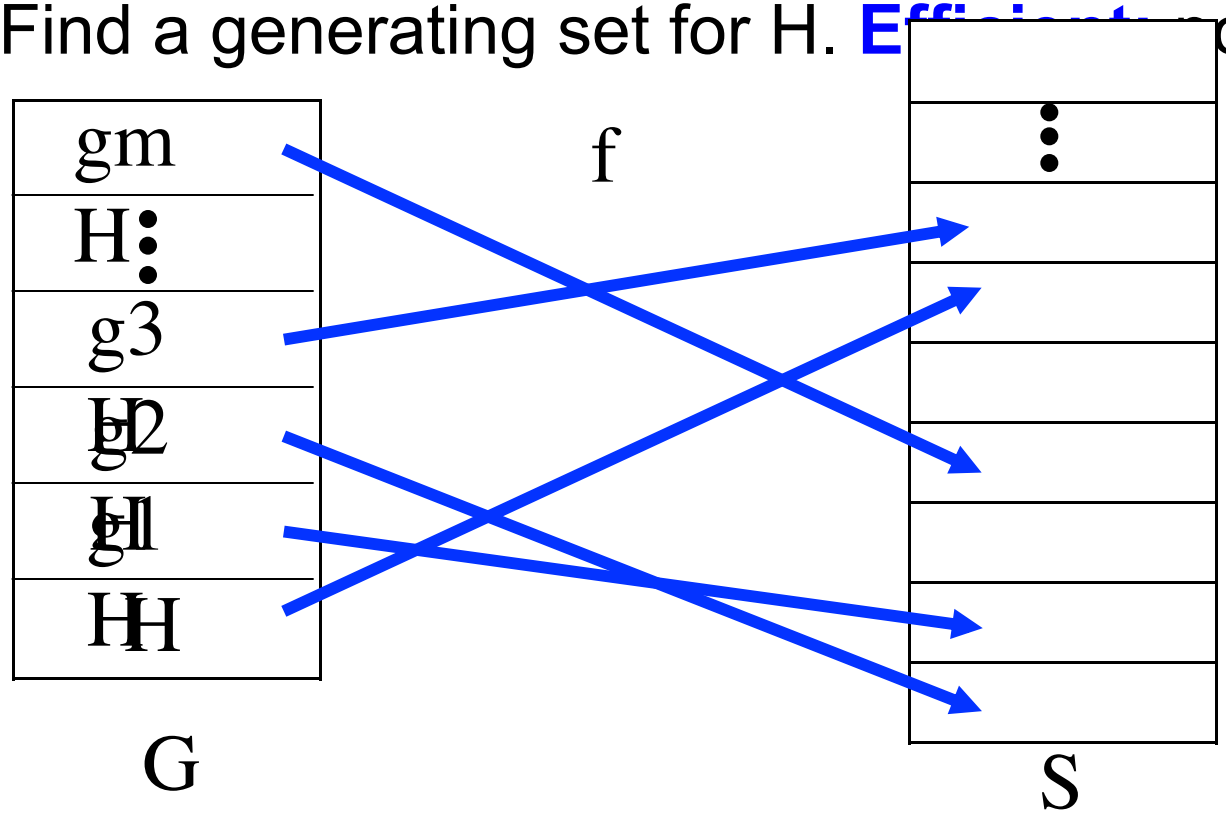


$$QFT = \frac{1}{\sqrt{2^n}} \sum_{x,p=0}^{2^n-1} \exp\left(\frac{2\pi i p x}{2^n}\right) |p\rangle\langle x|$$

Hidden Subgroup Problem

Setup: Fix a group G (known) and a subgroup $H \subseteq G$ (unknown). Given query access to a function $f : G \to S$ that
is constant and distinct on left cosets of H in G. We say that f hides H.

Problem: Find a generating set for H. **Efficient:** polylog(|G|)

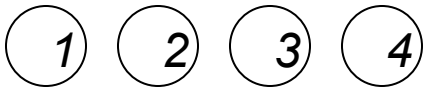


Hidden Subgroup Problems

Shor's factoring algorithm is an example of a HSP.

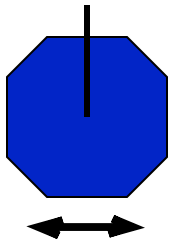
Efficient algorithms for the HSP over

the Symmetric group → an efficient algorithm for graph isomorphism



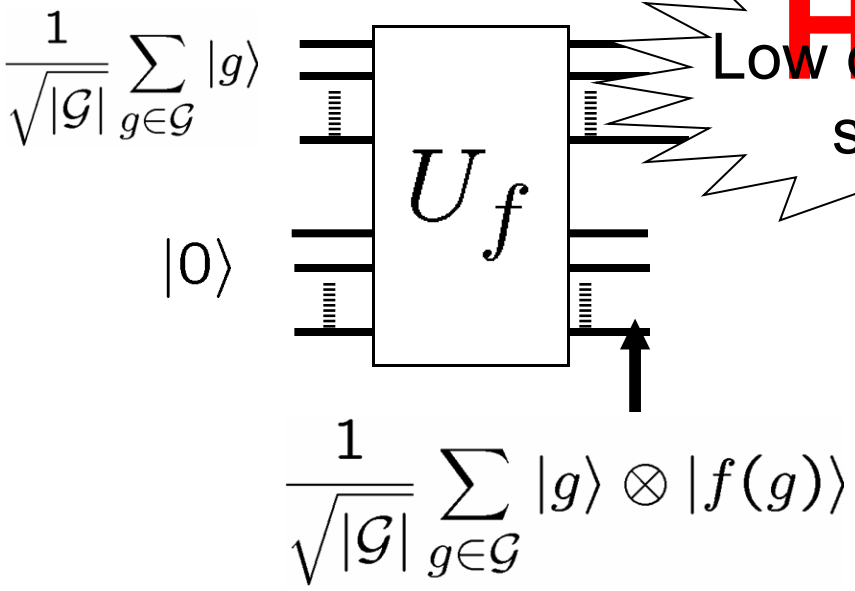
S_4

the Dihedral group → an efficient algorithm for poly(d)–unique shortest vector in a lattice



D8

Dihedral HSP

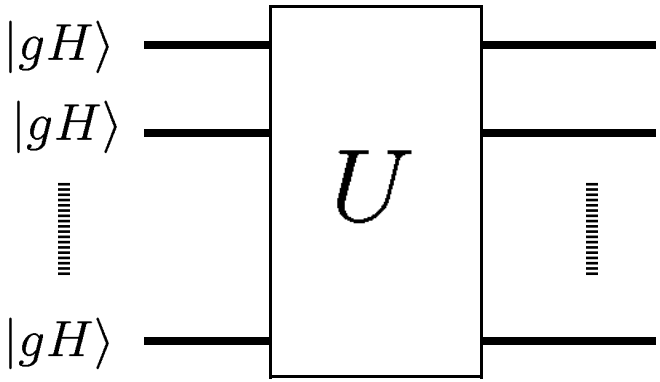


Low dimensional systems

$$|gH\rangle = \frac{1}{\sqrt{|H|}}\sum_{h\in H}|gh\rangle$$

Not much entanglement
AND
No way to make
an entangling measurement

k copies of the hidden subgroup states:



entangled measurement to determine the hidden subgroup

Query Complexity of HSP

Theorem: [Ettinger, Hoyer, Knill 99] k=O(log(N)) hidden subgroup states are sufficient to solve the hidden subgroup problem. (but no efficient quantum circuit is known.)

Theorem: [Bacon, Childs, van Dam 05] k=log(N) hidden subgroup states are necessary to solve the dihedral hidden subgroup problem.

1. Measurement across k copies is in entangled basis
2. Success probability of measurement is related to

The Subset Sum Problem

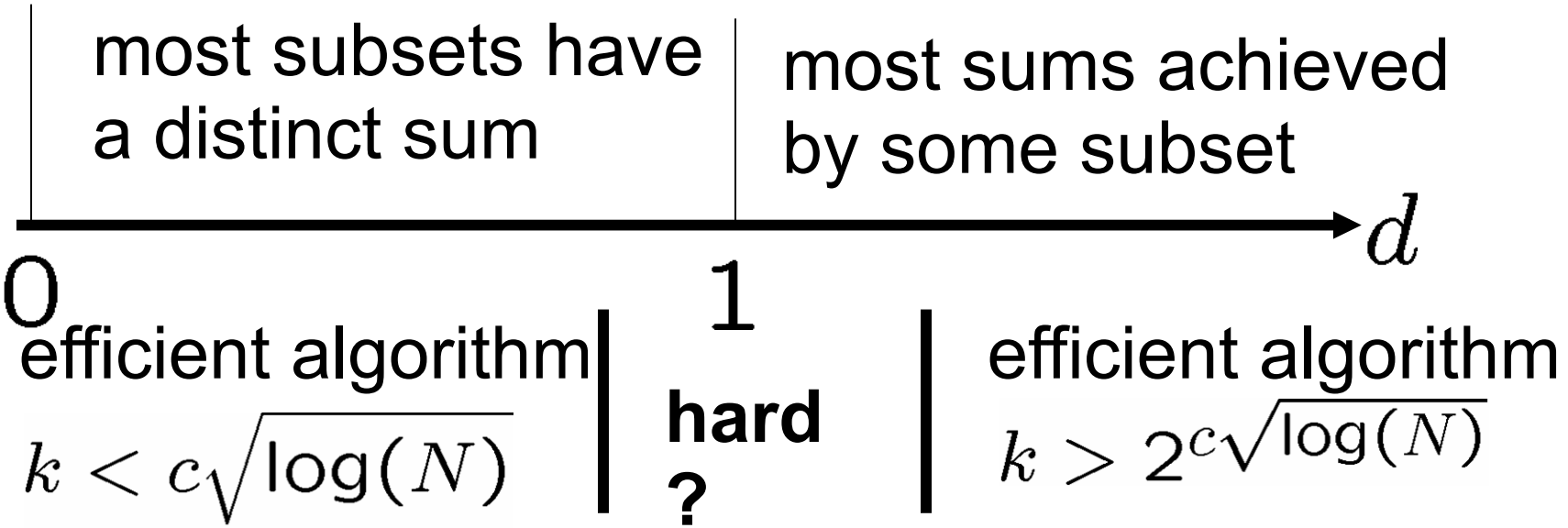
Given: k integers from $\mathbb{Z}_N := \{0, 1, \ldots, N-1\}$

i.e. $x \in \mathbb{Z}_N^k$, and a target $t \in \mathbb{Z}_N$

Problem: Does there exist a subset which sums to t

Define: density $d = \frac{k}{\log(N)}$

"low density" **"high density"**



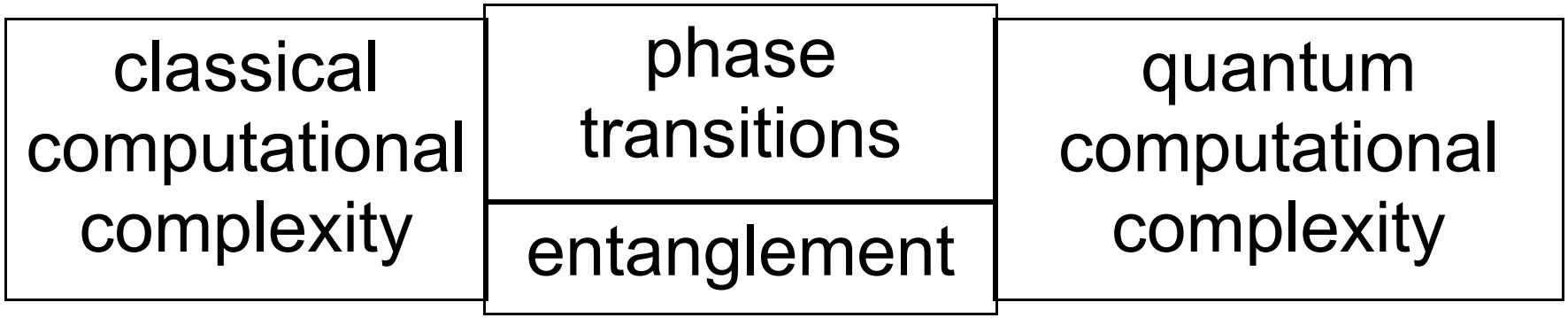
[Lagarias, Odlyzko

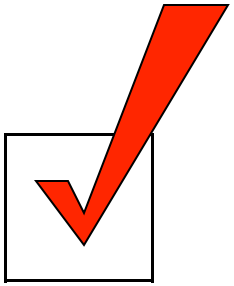
[Flaxman, Pryzdatek 04]

Query Complexity of HSP

Theorem: [Bacon, Childs, van Dam 05] k=log(N) hidden subgroup states are necessary to solve the dihedral hidden subgroup problem.

Phase transition in a classical compute science problem appearing in a quantum setting….



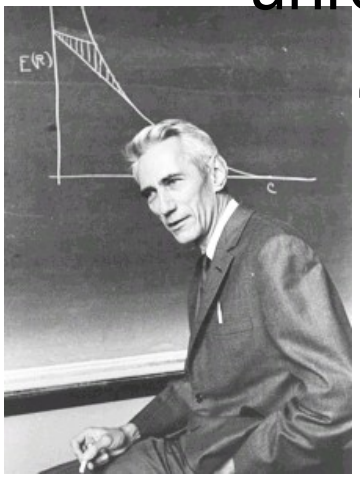


What gives quantum computers their **power** over classical computers

How to jumpstart the **building** of a quantum computer?

To Build a ~~Quantum~~ Computer Classical

Problem (1938): How to build a reliable system from unreliable components?



Claude Shannon

"A mathematical theory of communication" (1948)

How to communicate in the presence of noise

"Probabilistic logics and the synthesis of reliable organisms from unreliable components" (1956)

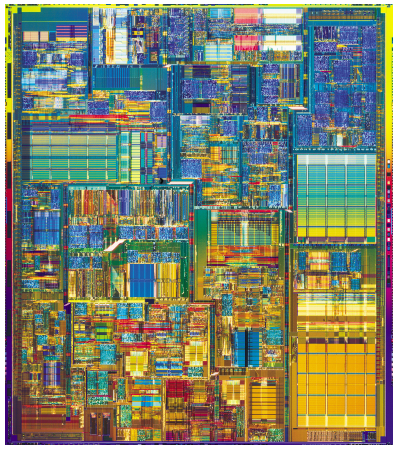


John von Neumann

How to compute when the subsystems of your computer can fail?

No Need for Theory?

Integrated Circuit



Hard Drive



The physics of these devices enacts the methods of the theory (von Neumann and Shannon)!

Two Paths

Hard Drive

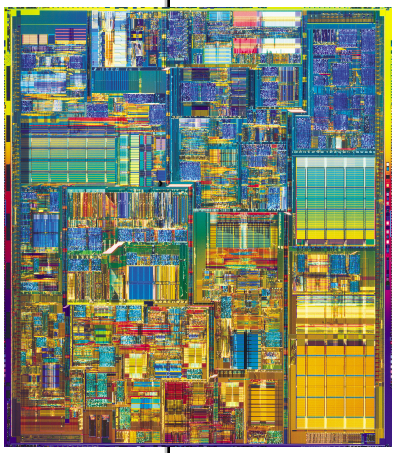


Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerance: macroscopic applied field fixed by local energy minimziation

Integrated Circuit



Coding: majority vote of current

Error correction: amplification

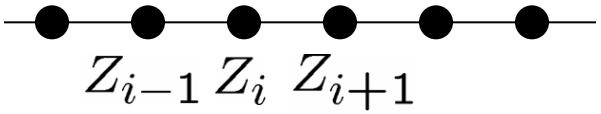
Fault-tolerance: conducting / insulating phase transition

The Physics Guara

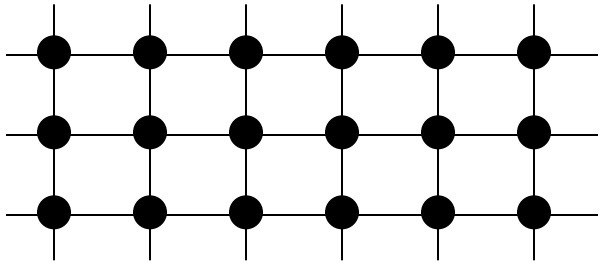
Classical **Ising model**: $E = -J \sum_{\langle i,j \rangle} Z_i Z_j \qquad Z_i \in \{+1, -1\}$

energy:=

one dimension



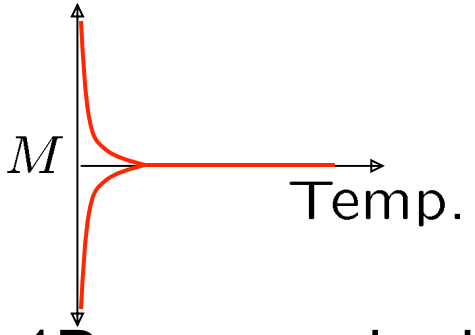
two dimensions



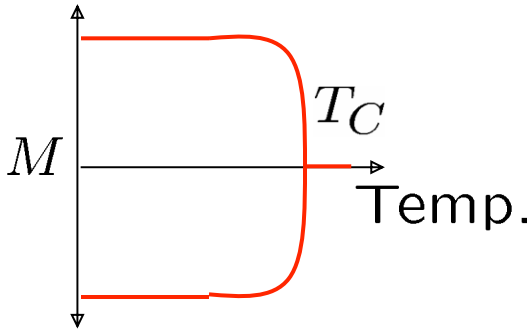
Redundancy code: $|0 \cdots 0\rangle,\ |1 \cdots 1\rangle$

Order parameter: total magnetization $M = \sum_i Z_i$

Information “still there”

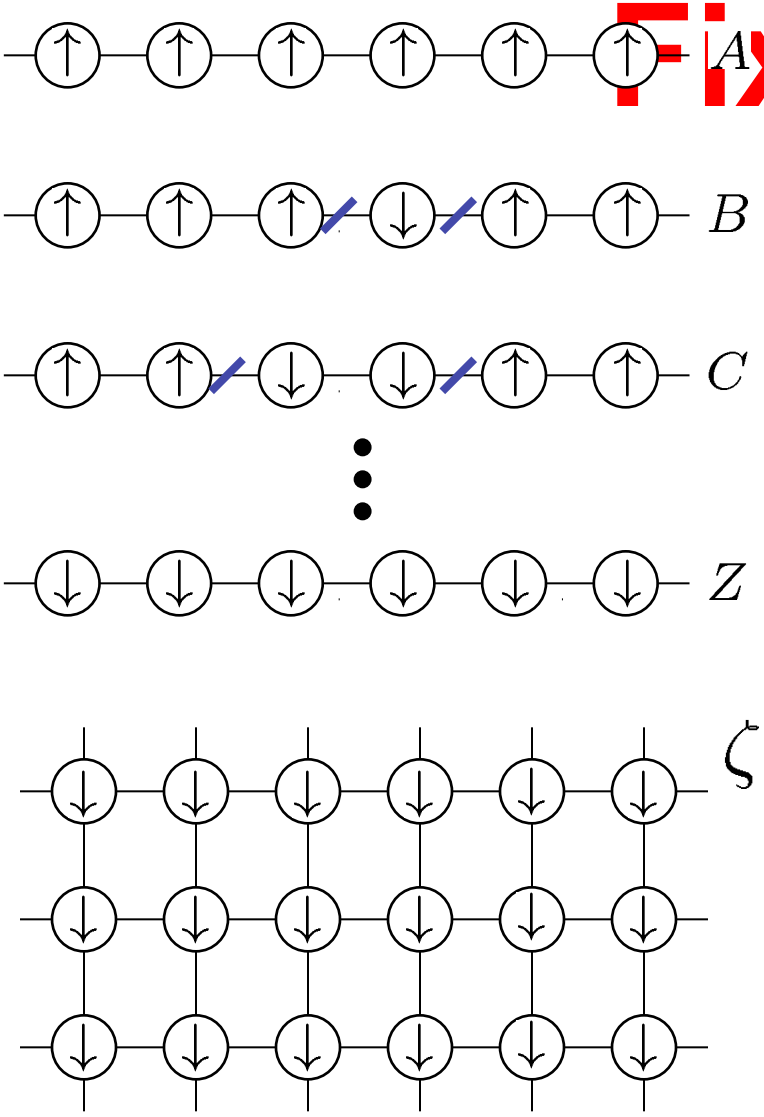


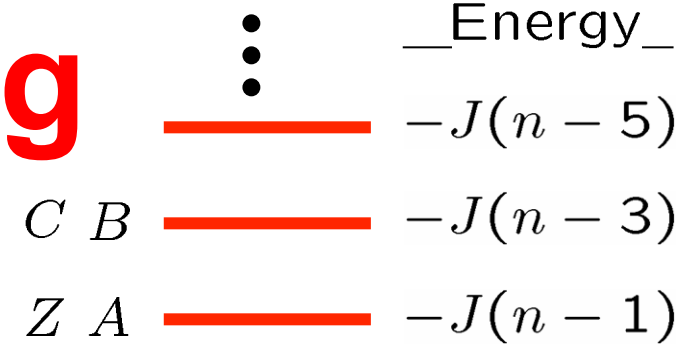
1D memory bad



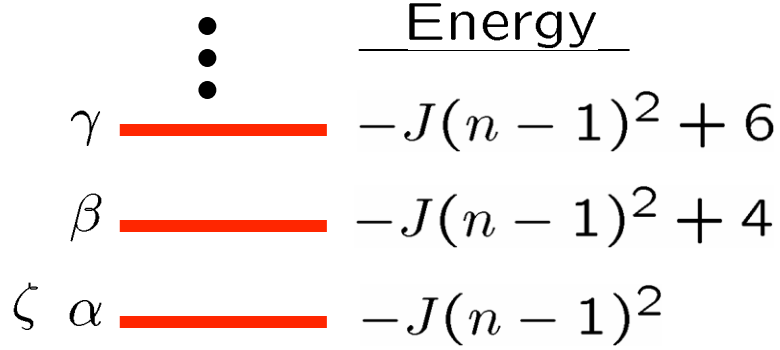
2D memory

Self Fixing





Excitations point-like
disordering entropy >> energy

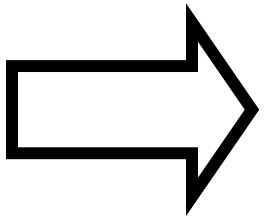


Excitations string-like
Energy grows like perimeter
disordering entropy << energy

To Build a Quantum Computer

Quantum Computer: Information is now quantum.

probabilistic bit:
p% 0, 1-p% 1



qubit:

$$\alpha|0\rangle + \beta|1\rangle, \alpha, \beta \in \mathbb{C}$$

Evolution is now quantum

Schrödinger equation, i.e. unitary evolution

$$|\psi_{out}\rangle = U|\psi_{in}\rangle \quad UU^\dagger = I$$

Measurement is now quantum

Probability of observing bitstring $|b\rangle$ is $|\langle b|\psi\rangle|^2$

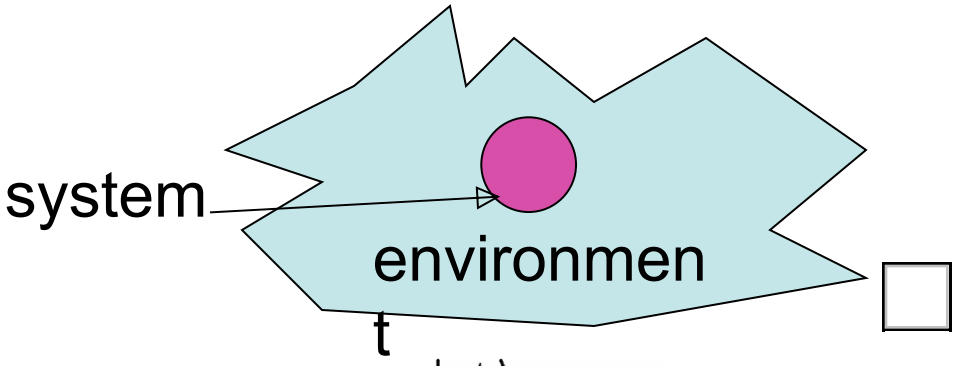
This description is very "analog" (compare "digital") looking?

Are quantum computers a reasonable computational model?

Even Worse!

Problems:

Real quantum systems are open quantum systems



$$|\psi\rangle \longrightarrow \rho$$

pure state $\longrightarrow$ mixed state

$$H = H_{sys} + H_{env} + H_{int}$$

system evolution

environment evolution

system-environment evolution

"system decoheres"

system evolution
is no longer unitary!

Real quantum experiments have finite precision

Perfect unitary evolution is a fiction.
Measurements are faulty. Preparation is imperfect.

Solutions

Observations

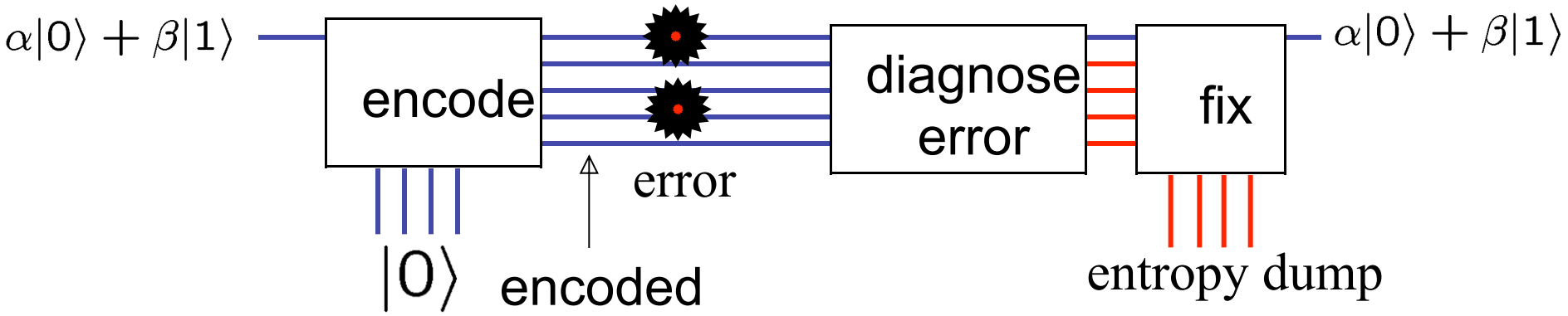
:

Quantum information can be encoded

$\alpha|0_L\rangle + \beta|1_L\rangle, \alpha, \beta \in \mathbb{C}$

$|0_L\rangle, |1_L\rangle$ span a subspace of some Hilbert space $\mathbb{C}^d$

Quantum error correcting codes



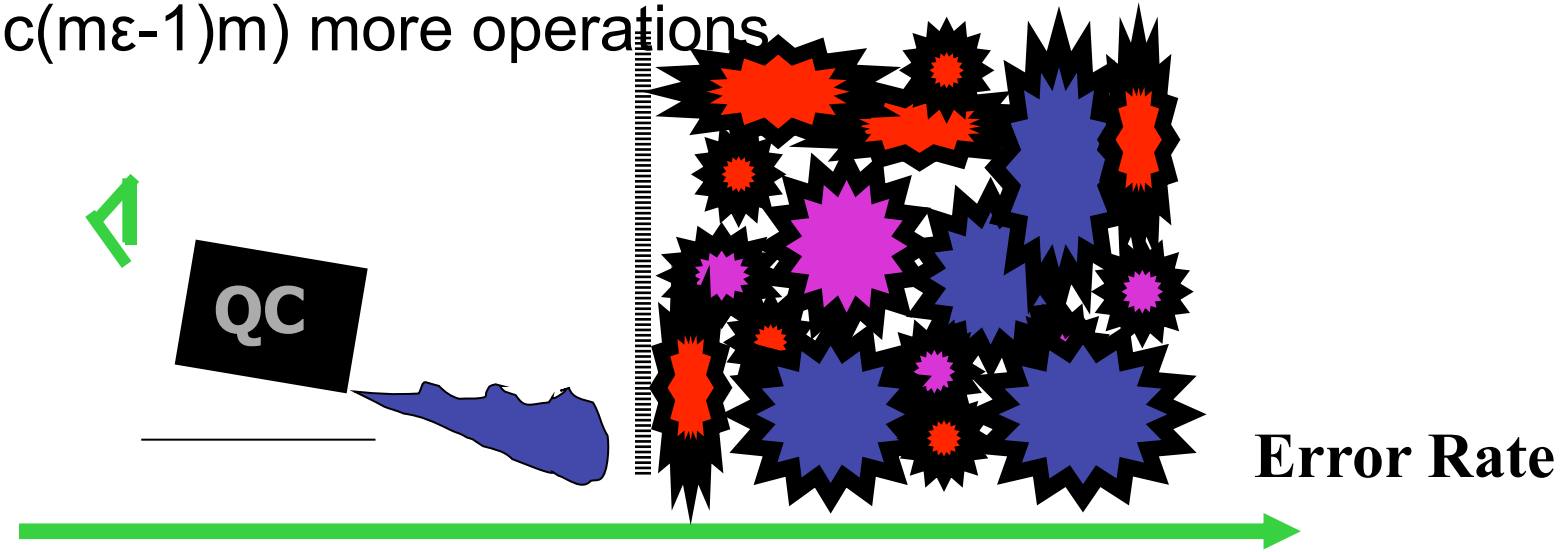
Quantum equivalent to Shannon's noisy coding

Quantum Threshold Theorem

Threshold Theorem:

[Aharanov, Ben-Or, Shor, Zurek, Knill, Laflamme, Kitaev, and Gottesman (95-96)]

If the error rate (decoherence or failure rate per operation) is smaller than some *pthresh*, then quantum computations of length
m can be enacted to accuracy ε at a cost of using a factor O(logc(mε-1)m) more operations.



Quantum computation is digital and is resilient to noise!

Whence Physics?

RANT MODE ON

Today: similar situation to early days of classical computation (threshold theorems but no physics!)

What is the phase of matter corresponding to the computer?

There are distinct PHYSICAL and DYNAMICAL reasons why robust classical computation is possible.

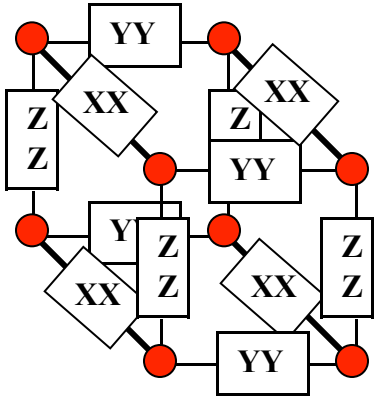
not all physical systems are equally good for computation

there exists systems whose <u>PHYSICS</u> guarantees their ability to enact robust classical computation.

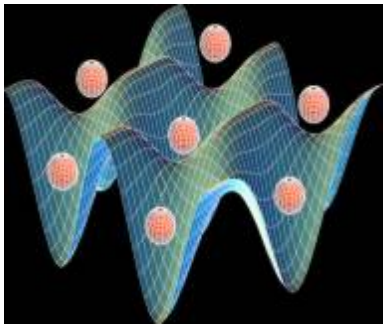
THE BILLION DOLLAR QUANTUM QUESTION:

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

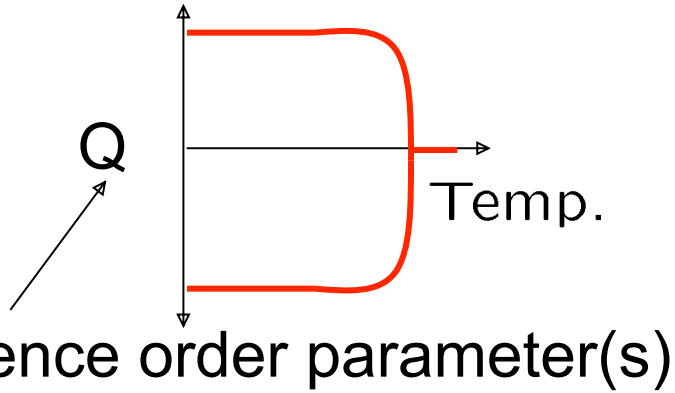
Self-Correcting Quantum Computers



Quantum many-body systems which have excitations which are string-like and are self-correct, but into which we can encode quantum information?



optical lattice



[Bacon, Ph.D. thesis, U.C. Berkeley 2001]
[Bacon, "Quantum Error Correcting Subsystems" in preparation]

Two Paths Diverged…



Traditional

Build a quantum computer
1 qubit at a time, with a severe device complexity overhead in the micro-architecture necessary for quantum error correction and fault-tolerance

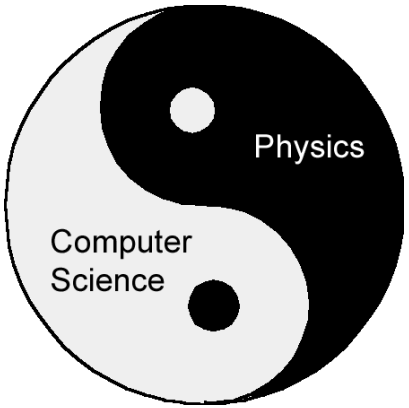
Self-Correcting

Engineer a many-body system whose physics enacts the quantum error correction and fault-tolerance

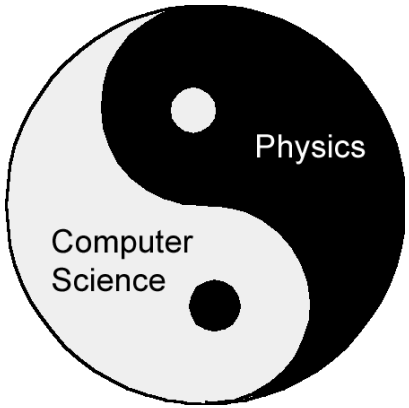
How and What to Quantum Compute

What gives quantum computers their **power** over classical computers

How to jumpstart the **building** of a quantum computer?



Dave Bacon
Santa Fe Institute



Collaborators



Univ. of Waterloo
Richard Cleve

Univ. of Toronto
Daniel Lidar

MIT
Isaac Chaung
Kenneth Brown
Aram Harrow

IBM T.J. Watson
David Divencenzo
Guido Burkard

U.C. Berkeley
K. Birgitta Whaley

Stanford
Xinlan Zhou

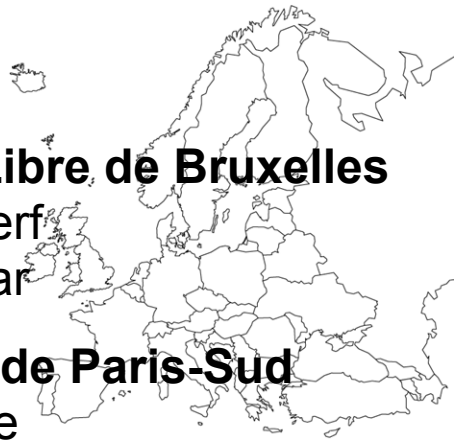
U.C. Santa Barbara
Wim van Dam

Univ. of New Mexico
Carl Caves
Ivan Deutsch
Brian Eastin
Tracey Tessier

Caltech
Andrew Childs
Debbie Leung
Benjamin Toner



Univ. of Queensland
Michael Nielsen
Michael Bremner
Jennifer Dodd

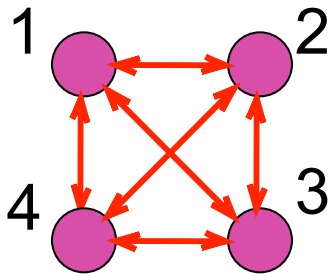


Universit Libre de Bruxelles
Nicholas Cerf
Serge Masar

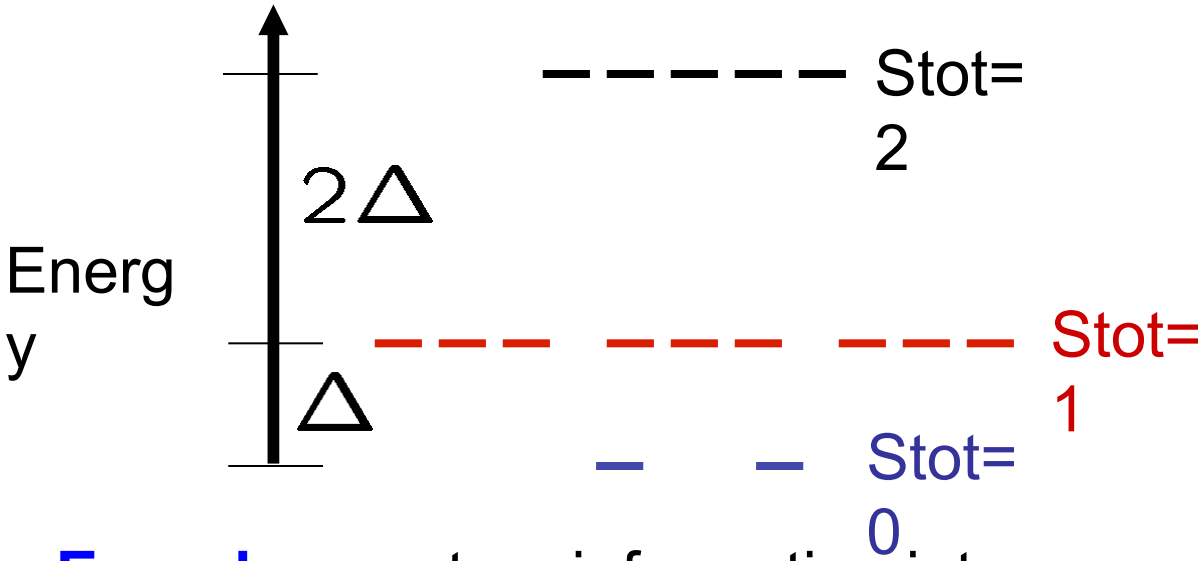
Universite de Paris-Sud
Julia Kempe

Exchange Everywhere

Can we engineer systems which turn dephasing/cooling → heating?



$$H = \Delta \sum_{i>j=1}^{4} \vec{S}_i \cdot \vec{S}_j = \frac{\Delta}{2}\left(S_{tot}^2 - 3I\right)$$



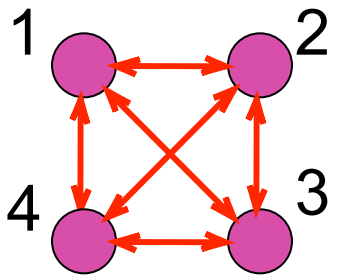
Encode quantum information into ground state
ground state is a quantum error detecting code

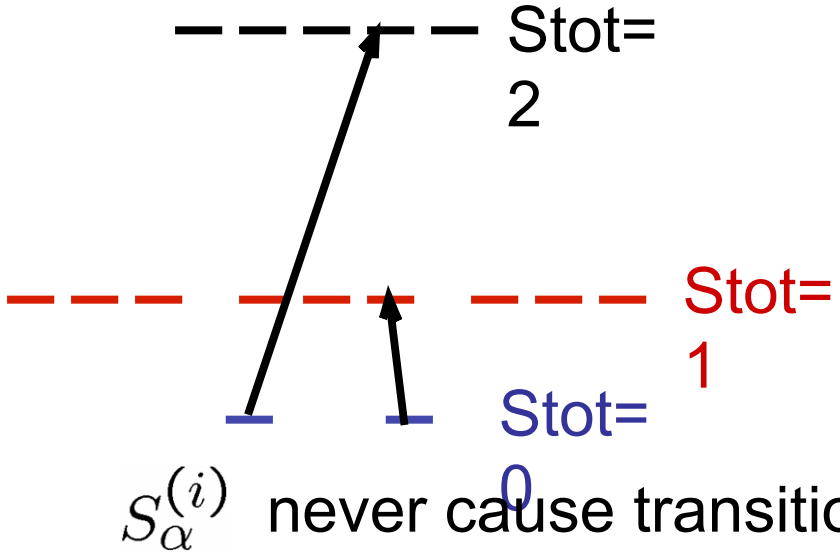
Supercoherence

[Bacon, Brown, and Whaley, *Phys. Rev. Lett.* 87, 247902 (2001)]

Independent coupling of system to environment

$$H_{int} = \sum_{i=1}^{4} S_\alpha^{(i)} \otimes B_\alpha^{(i)}$$





all single qubit interactions on this system are of heating

$S_\alpha^{(i)}$ never cause transitions on encoded space

All first order decoherence processes suppressed $kT \ll \Delta$